Practice Quiz 08D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 08

Restricting a domain to obtain an inverse.

1. Let $F : \mathbf{R} \to [2, \infty)$ be given by $F(x) = x^2 + 2$.

   (a) Show that F is not one-to-one.

   Solution The distinct real numbers 1 and -1 satisfy $F(1) = 3 = F(-1)$. Hence F is not one-to-one. ∎

   (b) Choose a subset $D \subseteq \mathbf{R}$ such that the restriction $F : D \to [2, \infty)$ has an inverse. Give the inverse and verify both compositions, with their domains stated.

   Solution Choose $D = [0, \infty)$. Define

   $$G : [2, \infty) \to [0, \infty), \qquad G(y) = \sqrt{y - 2}.$$

   For every $y \geq 2$, the square root is defined and nonnegative, and

   $$F(G(y)) = (\sqrt{y - 2})^2 + 2 = y.$$

   For every $x \geq 0$,

   $$G(F(x)) = \sqrt{(x^2 + 2) - 2} = \sqrt{x^2} = x,$$

   where the final equality uses $x \geq 0$. Thus $F \circ G$ is the identity on $[2, \infty)$ and $G \circ F$ is the identity on $[0, \infty)$. Consequently G is the inverse of the restricted function. ∎

Solutions revised using GPT-6 Astra in the Codex harness.